

Adjustable Spatio-Spectral Hyperspectral Image Compression Network

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Abstract—With the rapid growth of hyperspectral data archives in remote sensing, the need for efficient storage has become essential, driving significant attention toward learning-based hyperspectral image (HSI) compression. However, a comprehensive investigation of the individual and joint effects of spectral and spatial compression on learning-based HSI compression has not been thoroughly examined yet. Conducting such an analysis is crucial for understanding how the exploitation of spectral, spatial, and joint spatio-spectral redundancies affects HSI compression. To address this issue, in this article, we propose adjustable spatio-spectral hyperspectral image compression network (HyCASS), a learning-based model designed for adjustable HSI compression in both spectral and spatial dimensions. HyCASS consists of six main modules: spectral encoder module; spatial encoder module; compression ratio (CR) adapter encoder module; CR adapter decoder module; spatial decoder module; and spectral decoder module. The modules employ convolutional layers and transformer blocks to capture both short-range and long-range redundancies. Experimental results on three HSI benchmark datasets demonstrate the effectiveness of our proposed adjustable model compared to existing learning-based compression models, surpassing the state of the art by up to 2.36 dB in terms of peak signal-to-noise ratio. Based on our results, we establish a guideline for effectively balancing spectral and spatial compression across different CRs, taking into account the spatial resolution of the HSIs.

Index Terms—Adjustable spatio-spectral compression, deep learning, hyperspectral image (HSI) compression, remote sensing.

I. INTRODUCTION

HYPERSPECTRAL sensors capture images spanning hundreds of continuous bands across the electromagnetic spectrum. Fine spectral information provided in hyperspectral images (HSIs) enables the identification and differentiation of materials within a scene. Hyperspectral sensors, mounted on satellites, airplanes, and drones, enable a wide range of remote sensing (RS) applications [1], including forest monitoring [2], water quality assessment [3], wildfire detection [4], flood mapping [5], yield estimation [6], and plastic litter detection [7]. The continuous improvement in hyperspectral sensors has enabled them to extract increasingly detailed spectral and spatial information, which is essential for advanced analysis. However, the

substantial volume of data produced by these sensors presents significant challenges in terms of storage, transmission, and processing. An emerging research area focuses on the efficient compression of hyperspectral data to preserve crucial spectral and spatial information content for subsequent analysis [8].

Many HSI compression methods are presented in the literature. Generally, they can be categorized into the following three classes: lossless; near-lossless; and lossy HSI compression.

Each category offers a unique tradeoff between data preservation and compression efficiency. Lossless HSI compression ensures a perfect reconstruction of the original data without any loss of information and is, therefore, particularly important for tasks with zero tolerance for data degradation. However, lossless HSI compression methods typically only achieve compression ratios (CRs) of 2–4 [9]. This restriction limits their applicability in scenarios with strict bandwidth or storage space constraints that require significantly higher CRs. Near-lossless HSI compression achieves higher CRs than lossless compression while introducing minimal distortion between the original and reconstructed HSIs. The maximum error is upper bounded, ensuring controlled deviation in the reconstructed HSIs. However, small errors may accumulate, potentially impacting applications that require absolute spectral precision. Furthermore, the achievable CRs remain limited compared to lossy compression. Lossy HSI compression offers significant advantages, particularly in achieving high CRs, which is essential for applications with strict bandwidth or storage limitations [8]. By selectively discarding less important information, lossy compression efficiently reduces data size while preserving key features, making it ideal for large-scale hyperspectral data transmission, storage, and processing. Although lossy compression introduces some degree of information loss, the degradation is typically negligible and does not substantially compromise the usability of the data in most practical applications.

From a methodological point of view, HSI compression can be grouped into the following two categories: 1) traditional methods and 2) learning-based methods.

Traditional HSI compression methods have been extensively investigated in RS with predictive coding emerging as a common approach [10]. Predictive coding takes advantage of both spectral and spatial redundancies by predicting pixel values based on contextual information and encoding only the residuals between predicted and actual values for efficient storage or transmission. For example, in [11], a clustered differential pulse code modulation compression method, which clusters spectra and calculates an optimized predictor for each cluster, is proposed

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Data is available on-line at <https://git.tu-berlin.de/rsim/hycass>.

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for HSIs. After linear prediction of a spectrum, the difference is entropy-coded using an adaptive entropy coder for each of the clusters. Another prominent implementation of predictive coding is the CCSDS 123.0-B-2 standard [12] for lossless HSI compression that employs adaptive linear prediction to minimize redundancies in HSIs. Its low complexity and flexible architecture, coupled with the capability to achieve near-lossless compression through closed-loop quantization, make it well-suited for deployment in onboard RS systems. Although prediction-based methods are widely used for lossless and near-lossless HSI compression, they cannot generate meaningful latent representations, and their autoregressive functionality results in a slow processing speed.

In contrast, traditional transform coding methods excel at extracting compact latent representations from hyperspectral data and are frequently used for lossy compression. They project hyperspectral data into a lower dimensional, decorrelated latent space using mathematical transformations. The resulting reduced number of coefficients is subsequently quantized, introducing some loss of information, and then, entropy-coded. Several traditional transform-coding methods are proposed in RS. As an example, in [13], principle component analysis (PCA) is combined with JPEG2000 for joint spatio-spectral compression of HSIs, whereas PCA is applied followed by discrete cosine transform in [14]. In [15], 3-D transform coding is achieved by applying wavelet transformation in the spatial dimensions and Karhunen–Loève transform in the spectral dimension, followed by 3D-SPIHT for efficient lossy compression. In [16] 3D-SPECK, which takes advantage of the 3-D wavelet transform to efficiently encode HSIs by compressing the redundancies in all three dimensions, is introduced. Despite their effectiveness in extracting compact features and achieving high CRs, traditional methods often rely on hand-crafted transformations, which limit their ability to fully exploit the rich spatio-spectral structure of hyperspectral data. Consequently, these methods may not generalize well when applied to hyperspectral data captured under different conditions, sensor types, or scene characteristics.

To overcome these limitations, the development of learning-based HSI compression has recently attracted great attention in RS. Learning-based HSI compression methods leverage deep neural networks to automatically learn hierarchical and data-adaptive representations from large-scale training datasets, allowing more effective characterization of complex spectral and spatial redundancies. These methods also lead to an improved rate-distortion performance and better generalization across different scenes [8]. As an example, in [17], the LineRWKV method that enhances CCSDS 123.0-B-2 [12] by introducing a learning-based predictor is presented. LineRWKV achieves superior lossless and near-lossless reconstruction performance compared to CCSDS 123.0-B-2 at the cost of increased training time and computational complexity. Fuchs et al. [18] introduce two generative-adversarial-network-based models designed for spatio-spectral compression of HSIs. Their model extends the high fidelity compression framework [19] by incorporating: 1) squeeze and excitation blocks and 2) 3-D convolutions to better exploit spectral redundancies alongside spatial redundancies. Although these models are capable of achieving extremely high

CRs, exceeding 10^4 , their generative nature can lead to the synthesis of unrealistic spectral and spatial content, potentially compromising the fidelity of the reconstructed data. In [20], a channel clusterization strategy is proposed for onboard HSI compression to reduce the computational demands and make the method scalable for different data sources with a varying number of spectral bands. While the method achieves high quality reconstructions for low CRs, its robustness is highly dependent on a proper tuning of the number of features and the latent space dimensionality. In [21], a method for HSI compression using implicit neural representations is presented, where an multilayer perceptron (MLP) network learns to map pixel locations to pixel intensities. The weights of the learned model are stored or transmitted to achieve compression. In [22], a neural video representation approach is proposed. HSIs are treated as a stream of video data, where each spectral band represents a frame of information, and variances between spectral bands represent transformations between frames. Their approach utilizes the spectral band index and the spatial coordinate index as its input to perform network overfitting. A fundamental limitation of neural representation approaches lies in their training procedure, which involves overfitting a separate neural network to each HSI. This instance-specific optimization results in substantial computational costs, limiting the practicality of large-scale data processing.

Most state-of-the-art learning-based HSI compression methods adopt the autoencoder (AE) architecture, where an encoder network compresses the input data into a compact latent representation, and a decoder network reconstructs the data from that. This structure enables end-to-end optimization and allows the networks to learn efficient, data-driven representations of spectral and spatial information. Existing AE models differ primarily in how they process the spectral and spatial dimensions by using 1-D, 2-D, and 3-D convolutional layers, to balance compression efficiency, reconstruction quality, and computational complexity. As an example, in [23] and [24], 1-D convolutional autoencoder (1D-CAE) is presented, which compresses the spectral content without considering spatial redundancies by stacking multiple blocks of 1-D convolutions, pooling layers, and leaky rectified linear unit (LeakyReLU) activations. Although high-quality reconstructions can be achieved with this model, the pooling layers limit the achievable CRs to 2^n , where n is the number of poolings. Another limitation of this model is the increasing computational complexity with higher CRs due to the deeper network architecture. In [25], spectral signals compressor network (SSCNet) extracts spatial features via 2-D convolutional kernels. The 2-D max pooling layers introduce spatial compression, while the final CR is adapted via the latent channels within the bottleneck. Although SSCNet enables significantly higher CRs, this comes at the cost of spatially blurred image reconstructions. In [26], 3-D convolutional autoencoder (3D-CAE) is introduced for joint compression of spatio-spectral redundancies via 3-D convolutional kernels. In addition, residual blocks allow gradients to flow through the network directly and improve the model's performance. However, the 3-D kernels significantly increase computational complexity. In [27], spatio-spectral compression network (S2C-Net) is introduced.

Initially, a pixelwise AE is pretrained to capture the essential spectral features of the hyperspectral data. To enhance spatial redundancy removal, a spatial AE network is added to the bottleneck layer of the spectral AE. This dual-layer architecture allows the model to learn both spectral and spatial representations effectively. The entire model is then trained using a mixed loss function that combines reconstruction errors from both the spectral and spatial components. Although this model achieves state-of-the-art performance for high CRs, it falls short in optimally balancing the tradeoff between spectral and spatial CR. This limitation suggests room for improvement in achieving more balanced compression across both dimensions. In [28], the authors propose hyperspectral compression transformer (HyCoT), a transformer-based AE designed for pixelwise compression of HSIs that exploits long-range spectral redundancies. The model significantly reduces computational complexity through random sampling and independent pixelwise processing, without notable degradation in reconstruction quality. However, similar to other spectral compression models, HyCoT does not exploit spatial redundancies, thus limiting the achievable CRs.

Existing learning-based HSI compression models have limitations to effectively address several fundamental challenges associated with hyperspectral data, including varying spatial resolutions, high spectral dimensionality, and the need for adjustability across a broad range of CRs. In particular, these models often lack mechanisms to dynamically exploit spectral and spatial redundancies under specific compression requirements and sensor characteristics. As a result, current approaches face limitations in terms of scalability, generalization, and their ability to balance compression efficiency, reconstruction fidelity, and computational complexity.

To overcome the aforementioned critical issues, in this article, we introduce hyperspectral image compression network (HyCASS). The proposed model aims to enable flexible spatio-spectral compression. To this end, HyCASS employs six modules: a spectral encoder module, a spatial encoder module, a CR adapter encoder module, a CR adapter decoder module, a spatial decoder module, and a spectral decoder module.

The novelty of the proposed model consists of the following.

- 1) A spectral feature extraction that captures spectral redundancies across the whole spectrum of each pixel independently (realized within the spectral encoder module and spectral decoder module).
- 2) An adjustable number of spatial stages exploiting both short- and long-range spatial redundancies to control spatial compression (realized within the spatial encoder module and spatial decoder module).
- 3) An adjustable latent channel dimension to regulate spectral compression (realized within the CR adapter encoder module and CR adapter decoder module).

Unlike existing methods in the literature, our proposed model supports flexible compression in both spectral and spatial dimensions. We evaluate the proposed model on three distinct datasets (HySpecNet-11k [29], Berlin-Urban-Gradient [30], and MLRetSet [31]), demonstrating its effectiveness in compressing over a broad range of CRs and multiple spatial resolutions. Our experimental results show that spectral compression is

preferable in cases where the CR or spatial resolution is low, whereas spatio-spectral compression is more effective for medium and high CRs or spatial resolutions. The main contributions of this article are summarized as follows.

- 1) We propose a spatio-spectral HSI compression model with adjustable spatial stages and latent channels, capable of effective HSI compression across a broad range of CRs and varying spatial resolutions.
- 2) We provide extensive experimental results on three benchmark datasets for the introduced model, including an ablation study, comparisons with other approaches, and visual analyses of the reconstructions.
- 3) We analyze the effects of spectral and spatial HSI compression on the reconstruction quality for multiple CRs and three ground sample distances (GSDs), providing a comprehensive evaluation of their individual and combined impacts on reconstruction performance.
- 4) We demonstrate the advantages of adjustable deep learning architectures and derive guidelines for the tradeoff between spectral and spatial compression under varying CR conditions, for low, medium, and high spatial resolution hyperspectral data.

The rest of this article is organized as follows. Section II introduces the proposed HyCASS model. Section III describes the considered datasets and provides the design of experiments, while the experimental results are presented in Section IV. Finally, Section V concludes this article.

II. PROPOSED ADJUSTABLE SPATIO-SPECTRAL HYCASS

Let $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$ denote an HSI with spatial dimensions H and W , and C spectral bands. In this work, we focus on lossy HSI compression that transforms the original HSI $\mathbf{X}$ into a compact and decorrelated latent representation $\mathbf{Y} \in \mathbb{R}^{\Sigma \times \Omega \times \Gamma}$. Here, Σ and Ω denote the reduced latent spatial dimensions, while Γ represents the number of latent channels. The latent representation $\mathbf{Y}$ should retain sufficient information, such that the original HSI can be approximately reconstructed from $\mathbf{Y}$ as $\hat{\mathbf{X}} \in \mathbb{R}^{H \times W \times C}$. The compression aims to minimize the distortion $d : \mathbf{X} \times \hat{\mathbf{X}} \rightarrow [0, \infty)$ between $\mathbf{X}$ and $\hat{\mathbf{X}}$ for a fixed CR.

To enable effective spatio-spectral HSI compression, we propose HyCASS. The proposed model combines pixelwise convolutions, strided 2-D convolutional layers, and residual shifted windows (Swin) transformer blocks [32] to leverage both short-range and long-range redundancies across the spectral as well as the spatial dimension of HSIs. As illustrated in Fig. 1, HyCASS consists of the following six modules within both the encoder and decoder.

- 1) A spectral encoder module that involves a spectral feature extraction.
- 2) A spatial encoder module with a configurable number of spatial stages S for adjustable spatial compression, incorporating short-range and long-range spatial redundancies.
- 3) A CR adapter encoder module to balance the tradeoff between compressing spectral and spatial information content via the number of latent channels Γ , depending on the joint spatio-spectral target CR.

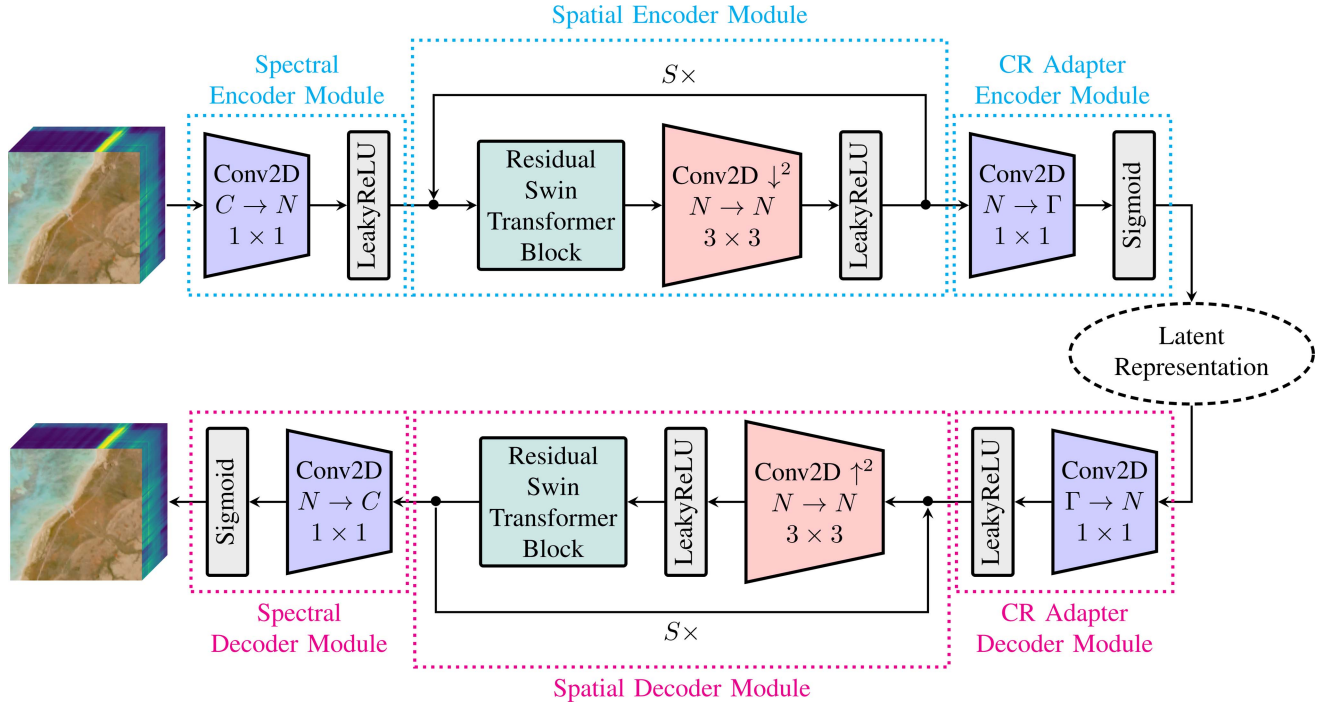


Fig. 1. Overview of our proposed HyCASS model. Initially, a pixelwise convolution in the spectral encoder module extracts spectral features. The spatial encoder module, composed of $S \times$ stacked stages, performs both long- and short-range spatial feature extraction, where each spatial stage introduces higher spatial compression. Subsequently, the CR adapter encoder module adjusts the size of the latent representation to match the targeted spatio-spectral CR. The decoder mirrors the encoder structure, replacing downsampling with upsampling operations.

- 4) A CR adapter decoder module that recovers spectral information.
- 5) A spatial decoder module that performs spatial reconstruction.
- 6) A spectral decoder module for spectral reconstruction.

In summary, the functional flow of HyCASS is as follows. First, the spectral encoder module extracts spectral redundancies, which are subsequently processed by the spatial encoder module to incorporate spatial redundancies. This joint representation is then further spectrally compressed by the CR adapter encoder module, forming a low-dimensional latent representation. On the decoder side, the latent representation is then spectrally expanded through the CR adapter decoder module, followed by a reconstruction of spatial information via the spatial decoder module. Finally, the spectral decoder module reconstructs spectral information.

HyCASS facilitates adjustable spatio-spectral compression, striking a balance between reconstruction fidelity and compression efficiency. This balance is achieved through the adjustable parameters Γ and S , which control spectral and spatial CR, respectively. In the following subsections, we provide detailed explanations of the six modules comprising HyCASS.

A. HyCASS Spectral Encoder Module

Initially, the spectral encoder module $E_\Phi : \mathbb{R}^{H \times W \times C} \rightarrow \mathbb{R}^{H \times W \times N}$ of the proposed HyCASS model, which is defined as

$$E_\Phi(\xi) = \text{LeakyReLU}(\text{Conv2D}_{1 \times 1}^{C \rightarrow N}(\xi)) \quad (1)$$

performs spectral feature extraction using a pixelwise convolution, realized via a 1×1 kernel inside a 2-D convolutional layer. Although 2-D convolutions are typically used for capturing spatial patterns, the use of a 1×1 kernel ensures that the convolution is applied independently to each pixel location without aggregating any spatial context. The pixelwise convolution captures spectral redundancies along the whole spectrum of each pixel by projecting the high-dimensional number C of spectral bands into a decorrelated, spectral representation with N channels. A LeakyReLU is applied after the convolutional layer to introduce nonlinear activation, thereby enhancing its capacity to learn complex patterns from data. This enables the spectral features extracted from the previous layer to capture more complex relationships in the hyperspectral data. The extracted spectral features are subsequently processed by the spatial encoder module.

B. HyCASS Spatial Encoder Module

Following the spectral encoding, HyCASS applies a spatial encoder module to capture and compress spatial redundancies within the hyperspectral data. The HyCASS spatial encoder $E_\chi : \mathbb{R}^{H \times W \times N} \rightarrow \mathbb{R}^{\Sigma \times \Omega \times N}$ defined as

$$\begin{cases} E_\chi(\xi) = f^{(S)}(\xi), & S \in \mathbb{N}_0 \\ f(\xi) = \text{LeakyReLU}(\text{Conv2D}_{3 \times 3}^{\downarrow 2}(\text{RSTB}(\xi))) \end{cases} \quad (2)$$

consists of a configurable sequence of spatial stages, denoted as S , each implemented as a neural transformation unit (NTU) [33].

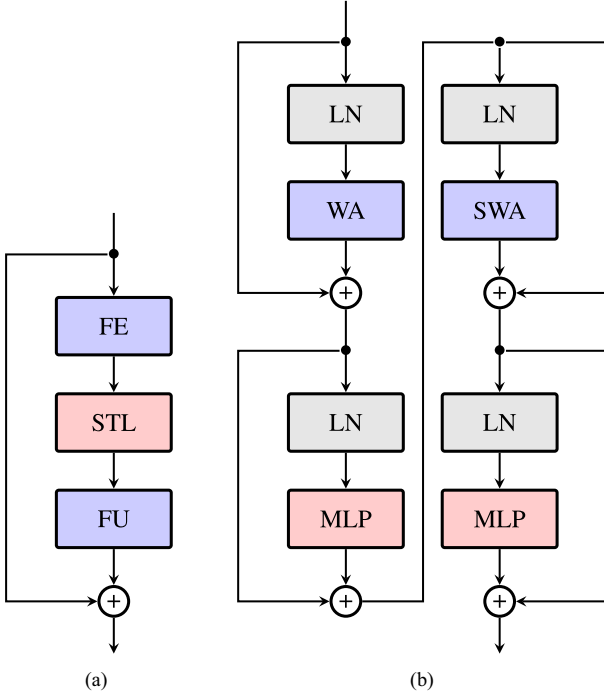


Fig. 2. Architecture of (a) RSTB and (b) STL. The RSTB captures long-range spatial redundancies using FE, STL, and FU subcomponents. The STL applies multihead self-attention within and across local windows using LN, WA, SWA, and MLP subcomponents. Layout is redesigned based on [32] and [33].

This systematically reduces the spatial dimension while enriching the feature representation with contextual spatial information. S is a configurable hyperparameter that determines the spatial CR, given by $\text{CR}_{\text{spat}} = 4^S$. This hyperparameter helps to adjust spatial compression to align with the targeted spatio-spectral CR. Each NTU performs stepwise spatial compression and is composed of the following three main components: a residual swin transformer block (RSTB) [32], a strided 2-D convolutional layer with a kernel size of 3×3 , and a LeakyReLU activation function.

The RSTB employs shifted window self-attention to capture long-range spatial redundancies as illustrated in Fig. 2.

It consists of the following three subcomponents.

- 1) Feature embedding (FE) that reorders the input feature channels into a token sequence.
- 2) Swin transformer layer (STL) that applies multi-head self-attention within and across local windows using layer normalization (LN), window attention (WA), shifted window attention (SWA), and MLP.
- 3) Feature unembedding (FU) that reorders the tokens back to their original spatial shape.

Residual connections mitigate the vanishing gradient issue, enhancing training stability. This design ensures efficient attention computation while preserving spatial locality and contextual information. Notably, patch division and linear embedding for tokenization are omitted as described in [33]. Following the RSTB, the strided 2-D convolution captures local short-range spatial redundancies. This layer downsamples the feature maps by a factor of 2 along both height and width, effectively reducing

the image size by a factor of 4 per stage. The number of channels N remains constant across all stages. A nonlinear LeakyReLU activation function is applied at the end of each NTU to enhance the model's capacity to learn complex patterns. This hierarchical spatial encoding progressively compresses the spatial content while preserving important structural and contextual details. We would like to note that in the case of zero spatial stages ($S = 0 \times$), HyCASS operates as spectral compression model while spatio-spectral compression is achieved with one or more spatial stages ($S > 0 \times$).

C. HyCASS CR Adapter Encoder Module

After capturing spectral and spatial redundancies using the respective encoder modules, HyCASS applies the CR adapter encoder module $E_{\Psi} : \mathbb{R}^{\Sigma \times \Omega \times N} \rightarrow \mathbb{R}^{\Sigma \times \Omega \times \Gamma}$ defined as

$$E_{\Psi}(\xi) = \text{Sigmoid}(\text{Conv2D}_{1 \times 1}^{N \rightarrow \Gamma}(\xi)). \quad (3)$$

It adjusts the number of latent channels Γ to fit the targeted spatio-spectral CR. Therefore, a 1×1 convolutional layer maps the spatio-spectral features from N channels to Γ latent channels. We use the sigmoid activation function to constrain the latent space to the range 0–1.

The spatio-spectral CR achieved by our proposed model arises from the joint contributions of both spectral and spatial compression. It can be expressed as

$$\text{CR} = \text{CR}_{\text{spec}} \cdot \text{CR}_{\text{spat}} = \frac{C}{\Gamma} \cdot 4^S \quad (4)$$

where $\text{CR}_{\text{spec}} = \frac{C}{\Gamma}$ denotes the spectral CR determined by the reduction of spectral channels from C to Γ , and $\text{CR}_{\text{spat}} = 4^S$ corresponds to the spatial CR introduced through S stages of spatial downsampling.

Although the spectral encoder module and the CR adapter encoder module are similarly structured, their roles differ: The spectral encoder module serves as an initial long-range spectral feature extractor, whereas the CR adapter encoder module's main purpose is to adjust the number of latent channels.

D. HyCASS CR Adapter Decoder Module

As the first step to perform reconstruction, the CR adapter decoder module $D_{\Psi'} : \mathbb{R}^{\Sigma \times \Omega \times \Gamma} \rightarrow \mathbb{R}^{\Sigma \times \Omega \times N}$ of HyCASS defined as

$$D_{\Psi'}(\xi) = \text{LeakyReLU}(\text{Conv2D}_{1 \times 1}^{\Gamma \rightarrow N}(\xi)) \quad (5)$$

applies a 1×1 convolution that projects the channels from Γ back to N , to match the NTU dimension, followed by a LeakyReLU.

E. HyCASS Spatial Decoder Module

Afterwards, the spatial decoder module $D_{\chi'} : \mathbb{R}^{\Sigma \times \Omega \times N} \rightarrow \mathbb{R}^{H \times W \times N}$ of HyCASS defined as

$$\begin{cases} D_{\chi'}(\xi) = g^{(S)}(\xi), & S \in \mathbb{N}_0 \\ g(\xi) = \text{RSTB}(\text{LeakyReLU}(\text{Conv2D}_{3 \times 3}^{\uparrow 2}(\xi))) \end{cases} \quad (6)$$

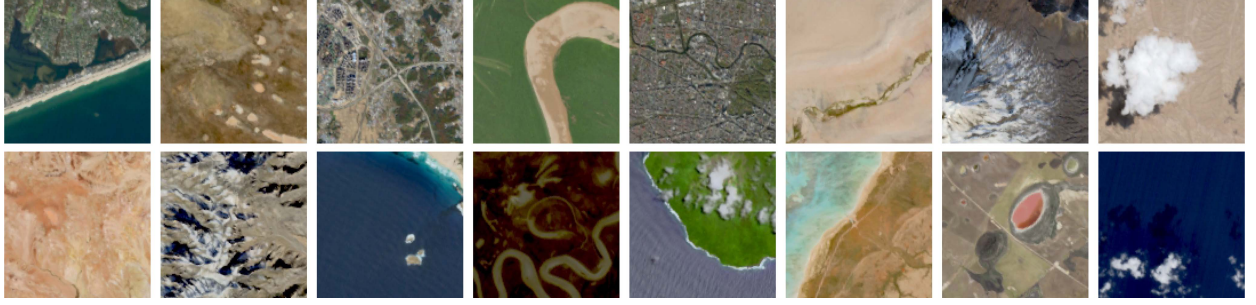


Fig. 3. Example of HSIs present in the HySpecNet-11 k dataset [29].

applies S stacked NTUs like the HyCASS spatial encoder module. However, in each NTU, first the 2-D convolution is applied to aggregate short-range spatial features and upsample the spatial dimensions. Then, the RSTB is applied for long-range spatial feature extraction.

F. HyCASS Spectral Decoder Module

Finally, the HyCASS spectral decoder module $D_{\Phi'}$: $\mathbb{R}^{H \times W \times N} \rightarrow \mathbb{R}^{H \times W \times C}$ defined as

$$D_{\Phi'}(\xi) = \text{Sigmoid}(\text{Conv2D}_{1 \times 1}^{N \rightarrow C}(\xi)) \quad (7)$$

projects the N channels back to the original C spectral bands using a 2-D convolution with a 1×1 kernel. A sigmoid activation constrains the reconstructed output intensities to the valid range of 0–1.

III. DATASET DESCRIPTION AND EXPERIMENTAL SETUP

A. Dataset Description

Three HSI datasets were employed in our experiments. These datasets differ in several aspects, including spatial resolution, number of spectral bands, and dataset size, summarized as follows.

1) *HySpecNet-11 k*: HySpecNet-11k [29] is a large-scale hyperspectral benchmark dataset constructed from 250 tiles acquired by the Environmental Mapping and Analysis Program (EnMAP) satellite [34]. It includes 11 483 nonoverlapping HSIs, each of which consists of 128×128 pixels and 202 spectral bands with a GSD of 30 m (low spatial resolution) and a spectral range of 420–2450 nm. The data are radiometrically, geometrically, and atmospherically corrected (i.e., the L2A water and land product). We used the recommended splits from [29] for training, validation, and test sets covering 70 %, 20 %, and 10 % of the HSIs, respectively. Fig. 3 illustrates example images from this dataset. We would like to note that we used HySpecNet-11k to show the effectiveness and generalization capability of our proposed model, particularly in the context of large-scale training with low spatial resolution HSIs.

2) *Berlin-Urban-Gradient*: Berlin-Urban-Gradient [30] is a hyperspectral dataset covering a region along the urban–rural gradient of Berlin. It comprises an airborne scene captured by the HyMap sensor [35] with 3.6 m GSD (medium spatial resolution) and a spectral range between 450 and 2 500 nm, which was

acquired on 20 August 2009 under clear sky conditions. The tile was split into 160 nonoverlapping HSIs of size 80×80 pixels with 111 spectral bands each. We split the data into the following:

- 1) a training set that includes 70 % of the HSIs;
- 2) a validation set that includes 20 % of the HSIs;
- 3) test set that includes 10 % of the HSIs.

Fig. 4 shows exemplary images of this dataset. We would like to note that we used the Berlin-Urban-Gradient dataset to show the effectiveness of HyCASS on HSIs with a medium spatial resolution and only few training samples.

3) *MLRetSet*: MLRetSet [31] is a hyperspectral benchmark dataset created from high spatial resolution hyperspectral imagery with 27.86 cm GSD. The hyperspectral dataset was acquired during an airborne flight covering the Turkish towns Yenice and Yeşilkaya on 4 May 2019. The 12 acquired tiles were split into 3 840 nonoverlapping HSIs of size 100×100 pixels with 369 spectral bands each. We split the data into the following:

- 1) a training set that includes 70 % of the HSIs;
- 2) a validation set that includes 20 % of the HSIs;
- 3) test set that includes 10 % of the HSIs.

Fig. 5 provides visual examples of typical scenes present in the MLRetSet dataset. We would like to note that we employed the MLRetSet dataset to demonstrate the effectiveness of our proposed model on HSIs with a high spatial resolution.

B. Experimental Setup

Our code was implemented in PyTorch based on the CompressAI [36] framework. In the case of MLRetSet, the HSIs were center-cropped to 96×96 pixels to facilitate repeated spatial downsampling by factors of two, which is required for spatial compression models. For HySpecNet-11k, we followed the easy split as introduced in [29]. Training runs were carried out on a single NVIDIA A100 SXM4 80 GB GPU using the Adam optimizer [37]. For the loss function, we employed the mean squared error (MSE) between the original HSI $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$ and its reconstruction $\hat{\mathbf{X}} \in \mathbb{R}^{H \times W \times C}$ as follows:

$$\text{MSE}(\mathbf{X}, \hat{\mathbf{X}}) = \frac{1}{H \cdot W \cdot C} \sum_{h,w,c} (\mathbf{X}(h, w, c) - \hat{\mathbf{X}}(h, w, c))^2 \quad (8)$$

where H , W , and C denote HSI height, width, and spectral bands, respectively, while $\mathbf{X}(h, w, c)$ represents the value of spectral band c at pixel location (h, w) .



Fig. 4. Example of HSIs present in Berlin-Urban-Gradient dataset [30].

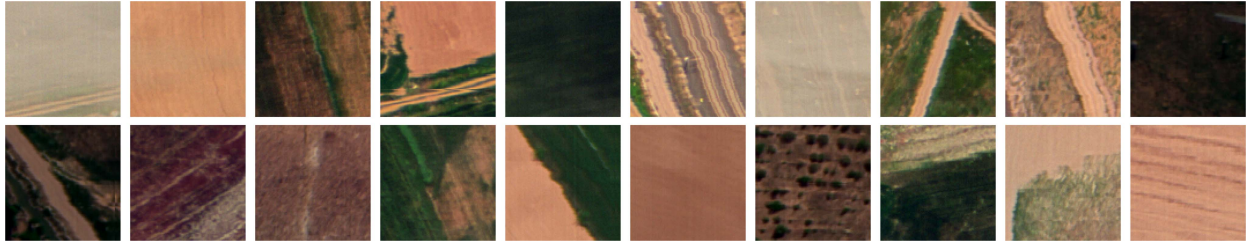


Fig. 5. Example of HSIs present in the MLRetSet dataset [31].

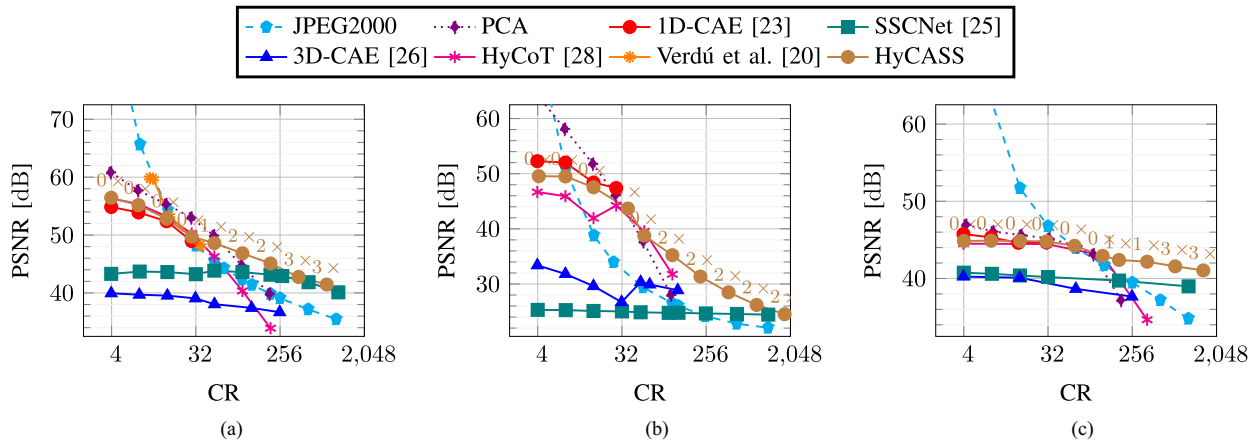


Fig. 6. Rate-distortion performance on the test set of (a) HySpecNet-11k [29] (easy split), (b) Berlin-Urban-Gradient [30], and (c) MLRetSet [31]. Rate is visualized as CR and distortion is given as PSNR in decibels (dB). (a) HySpecNet-11k (easy split). (b) Berlin-Urban-Gradient. (c) MLRetSet.

TABLE I
VALUES OF TRAINING HYPERPARAMETER SELECTED FOR EACH MODEL FOR THE HYSPECNET-11K [29], BERLIN-URBAN-GRADIENT [30], AND MLRETSET [31] DATASET

Model	HySpecNet-11k [29]			Berlin-Urban-Gradient [30]			MLRetSet [31]		
	Epochs	LR	BS	Epochs	LR	BS	Epochs	LR	BS
1D-CAE [23]	500	10^{-4}	2	500	10^{-4}	2	100	10^{-4}	2
SSCNet [25]	2,000	10^{-5}	8	2,000	10^{-5}	8	400	10^{-5}	8
3D-CAE [26]	1,000	10^{-4}	2	1,000	10^{-4}	2	250	10^{-4}	2
HyCoT [28]	2,000	10^{-3}	64	2,000	10^{-3}	64	500	10^{-3}	64
HyCASS	200	10^{-4}	16	1,000	10^{-4}	16	100	10^{-4}	16

We compare our proposed model with two traditional compression methods (JPEG2000 and PCA) and the following learning-based models:

- 1) 1D-CAE [23], a 1-D convolutional AE performing spectral compression;
- 2) SSCNet [25], a 2-D convolutional AE performing spatial compression;
- 3) 3D-CAE [26], a 3-D convolutional AE jointly compressing spatial and spectral redundancies;
- 4) Verdú et al. [20], a 2-D variational AE based on channel clustering;

- 5) HyCoT [28], a transformer-based AE exploiting spectral redundancies.

Training epochs, learning rate (LR), and batch size (BS) were adjusted per model and dataset to optimize training time and GPU memory usage, while ensuring convergence of the loss function for each training run. To this end, the optimal hyperparameters were empirically determined through preliminary experiments (grid search) and analysis of the obtained results. Table I lists the specific training hyperparameters used for each model and dataset configuration. It is worth noting that on HySpecNet-11k for 1D-CAE, the number of epochs was

reduced to 250 for $CR \in \{8, 16, 32\}$ due to runtime limitations. For MLRetSet, we had to reduce the BS to 1 for 1D-CAE at $CR = 32$ due to GPU memory constraints. Furthermore, we used the random sampling strategy from [28] to efficiently train HyCoT.

For HyCASS, we fixed the number of spatial encoder module channels to $N = 128$ and the swin transformer's window size was set to 8, consistent with the configuration used in the transformer-based image compression model [33], and all remaining parameters were aligned accordingly. The number of spatial stages was varied between $0 \times$ and $3 \times$ to assess the effect of different spatial compression levels. We would like to note that for zero spatial stages, we increased N to 1024 to compensate for the missing spatial stages in terms of model parameters and floating point operations (FLOPs). Also, we applied the random sampling strategy from HyCoT [28] and increased the LR to 10^{-3} , the BS to 64, and the number of epochs to 2000 for HySpecNet-11 k and Berlin-Urban-Gradient and 1000 for MLRetSet. For the Berlin-Urban-Gradient dataset, we had to reduce the window size to 4 when using $3 \times$ spatial stages. It is important to note that while the targeted spatio-spectral CRs were $CR \in \{4, 8, 16, \dots, 1024\}$, the achieved CRs may deviate due to the spectral dimension not being divisible by powers of two.

C. Evaluation Metrics

For the evaluation of the HSI compression methods, we consider the following two kinds of metrics:

- 1) metrics that measure the compression efficiency;
- 2) metrics that measure the reconstruction quality.

In our experiments, we use the CR to quantify the data reduction. Peak signal-to-noise ratio (PSNR), spectral angle (SA), and structural similarity index measure (SSIM) are used to measure the fidelity of a reconstructed HSI. Given an original HSI $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, its latent representation $\mathbf{Y} \in \mathbb{R}^{\Sigma \times \Omega \times \Gamma}$ and reconstruction $\hat{\mathbf{X}} \in \mathbb{R}^{H \times W \times C}$, the metrics are defined as follows.

1) *Compression Ratio (CR)*: The CR between an original HSI $\mathbf{X}$ with bit depth N_b and its representation in the latent space $\mathbf{Y}$ with bit depth $\hat{N}_b$ after encoding can be expressed as follows:

$$CR(\mathbf{X}, \mathbf{Y}) = \frac{N_b \cdot H \cdot W \cdot C}{\hat{N}_b \cdot \Sigma \cdot \Omega \cdot \Gamma}. \quad (9)$$

A higher CR indicates greater compression of the hyperspectral data, which could result in increased loss of information during reconstruction.

2) *Peak Signal-to-Noise Ratio (PSNR)*: For measuring the reconstruction quality, we use the PSNR between original HSI $\mathbf{X}$ and reconstructed HSI $\hat{\mathbf{X}}$, which is defined as

$$PSNR(\mathbf{X}, \hat{\mathbf{X}}) = 10 \cdot \log_{10} \left(\frac{MAX^2}{MSE(\mathbf{X}, \hat{\mathbf{X}})} \right) \quad (10)$$

where MAX denotes the maximum possible pixel value (e.g., 1.0 in the case of min-max normalization), and the mean squared

error (MSE) is defined as in (8). A higher PSNR value indicates better reconstruction quality with less distortion.

3) *Spectral Angle (SA)*: We also report the SA defined as

$$SA(\mathbf{X}, \hat{\mathbf{X}}) = \frac{1}{H \cdot W} \sum_{h,w} \frac{180}{\pi} \arccos \left(\frac{\sum_c \mathbf{X}(h, w, c) \cdot \hat{\mathbf{X}}(h, w, c)}{\sqrt{\sum_c \mathbf{X}(h, w, c)^2} \sqrt{\sum_c \hat{\mathbf{X}}(h, w, c)^2}} \right) \quad (11)$$

which quantifies the average spectral similarity between all pixels of an original $\mathbf{X}$ and a reconstructed HSI $\hat{\mathbf{X}}$. A smaller SA indicates higher spectral similarity and is inherently scale invariant.

4) *Structural Similarity Index Measure (SSIM)*: To measure the perceived quality of a reconstruction, we use the SSIM defined as:

$$SSIM(\mathbf{X}, \hat{\mathbf{X}}) = \frac{(2\mu_X \mu_{\hat{X}} + c_1)(2\sigma_{X\hat{X}} + c_2)}{(\mu_X^2 + \mu_{\hat{X}}^2 + c_1)(\sigma_X^2 + \sigma_{\hat{X}}^2 + c_2)}. \quad (12)$$

where μ_X and $\mu_{\hat{X}}$ are the means, σ_X^2 and $\sigma_{\hat{X}}^2$ are the variances, and $\sigma_{X\hat{X}}$ is the covariance of $\mathbf{X}$ and $\hat{\mathbf{X}}$. c_1 and c_2 are constants to provide stability against weak denominators. The SSIM metric yields values ranging from -1 to 1 , with higher values indicating reconstructions of higher perceptual quality.

IV. EXPERIMENTAL RESULTS

We conducted three sets of experiments, aiming at the following:

- 1) assessment of the effects of spectral and spatial compression within the proposed HyCASS model through an ablation study on three benchmark datasets;
- 2) comparison of our model's effectiveness with traditional baselines and lossy learning-based state-of-the-art HSI compression models;
- 3) qualitative analysis of the reconstruction results.

A. Ablation Study

In this subsection, we analyze the impact of varying the spatial stages S and latent channels Γ inside HyCASS on the reconstruction quality for multiple CRs. To assess generalization across different spatial resolutions, we report results on the HySpecNet-11k, the Berlin-Urban-Gradient and the MLRetSet datasets. For each configuration, S defines the spatial CR (CR_{spat}), while Γ determines the spectral CR (CR_{spec}). Given CR_{spat} , CR_{spec} is adjusted accordingly to match the targeted spatio-spectral CR.

1) *HySpecNet-11 k*: Table II shows the results of the ablation study on the HySpecNet-11 k dataset. From the table, one can derive the following three key observations.

First, for $CR < 32$, HyCASS with zero spatial stages (i.e., spectral compression only) yields superior reconstruction performance compared to HyCASS with one or more spatial stages (i.e., spatio-spectral compression). For example, when $CR \approx 4$, HyCASS with zero spatial stages achieves a PSNR of 56.44 dB while reconstruction quality reduces for one, two, and

TABLE II

HYCASS RESULTS OBTAINED BY VARYING THE SPATIAL STAGES S ON THE EASY SPLIT TEST SET OF THE HYSPECNET-11k [29] DATASET

S	CR	CR _{spec}	CR _{spat}	PSNR $\uparrow$	SA $\downarrow$
0 $\times$	3.96	3.96	1.00	56.44 dB	1.39°
1 $\times$	3.96	0.99	4.00	49.78 dB	2.44 $^\circ$
2 $\times$	3.97	0.25	16.00	48.45 dB	2.65 $^\circ$
3 $\times$	3.97	0.06	64.00	44.65 dB	3.33 $^\circ$
0 $\times$	7.77	7.69	1.00	55.16 dB	1.56°
1 $\times$	7.77	1.94	4.00	49.83 dB	2.44 $^\circ$
2 $\times$	7.77	0.49	16.00	48.08 dB	2.71 $^\circ$
3 $\times$	7.77	0.12	64.00	44.79 dB	3.27 $^\circ$
0 $\times$	15.54	15.54	1.00	52.83 dB	1.84°
1 $\times$	15.54	3.88	4.00	50.39 dB	2.39 $^\circ$
2 $\times$	15.54	0.97	16.00	48.80 dB	2.64 $^\circ$
3 $\times$	15.54	0.24	64.00	46.06 dB	3.08 $^\circ$
0 $\times$	28.86	28.86	1.00	49.72 dB	2.27°
1 $\times$	28.86	7.21	4.00	49.25 dB	2.55 $^\circ$
2 $\times$	28.86	1.80	16.00	48.03 dB	2.72 $^\circ$
3 $\times$	28.86	0.45	64.00	44.94 dB	3.24 $^\circ$
0 $\times$	50.50	50.50	1.00	45.86 dB	3.15 $^\circ$
1 $\times$	50.50	12.63	4.00	48.61 dB	2.72 $^\circ$
2 $\times$	50.50	3.16	16.00	48.58 dB	2.64°
3 $\times$	50.50	0.79	64.00	45.92 dB	3.11 $^\circ$
0 $\times$	101.00	101.00	1.00	39.84 dB	5.52 $^\circ$
1 $\times$	101.00	25.25	4.00	45.97 dB	3.16 $^\circ$
2 $\times$	101.00	6.31	16.00	46.84 dB	2.91°
3 $\times$	101.00	1.58	64.00	44.44 dB	3.32 $^\circ$
0 $\times$	202.00	202.00	1.00	32.97 dB	12.51 $^\circ$
1 $\times$	202.00	50.50	4.00	43.35 dB	3.77 $^\circ$
2 $\times$	202.00	12.63	16.00	45.09 dB	3.16°
3 $\times$	202.00	3.16	64.00	44.14 dB	3.43 $^\circ$
1 $\times$	404.00	101.00	4.00	41.05 dB	4.55 $^\circ$
2 $\times$	404.00	25.25	16.00	42.65 dB	3.72 $^\circ$
3 $\times$	404.00	6.31	64.00	42.76 dB	3.65°
1 $\times$	808.00	202.00	4.00	36.39 dB	7.45 $^\circ$
2 $\times$	808.00	50.50	16.00	40.81 dB	4.22 $^\circ$
3 $\times$	808.00	12.63	64.00	41.46 dB	3.95°

TABLE III

HYCASS RESULTS OBTAINED BY VARYING THE SPATIAL STAGES S ON THE TEST SET OF THE BERLIN-URBAN-GRADIENT [30] DATASET

S	CR	CR _{spec}	CR _{spat}	PSNR $\uparrow$	SA $\downarrow$
0 $\times$	4.11	4.11	1.00	49.55 dB	1.16°
1 $\times$	4.11	1.03	4.00	41.99 dB	2.39 $^\circ$
2 $\times$	4.11	0.26	16.00	38.67 dB	3.44 $^\circ$
3 $\times$	4.11	0.06	64.00	29.24 dB	6.48 $^\circ$
0 $\times$	7.93	7.93	1.00	49.50 dB	1.18°
1 $\times$	7.93	1.98	4.00	41.78 dB	2.45 $^\circ$
2 $\times$	7.93	0.50	16.00	37.80 dB	3.78 $^\circ$
3 $\times$	7.93	0.12	64.00	25.50 dB	10.45 $^\circ$
0 $\times$	15.86	15.86	1.00	47.53 dB	1.45°
1 $\times$	15.86	3.96	4.00	41.84 dB	2.42 $^\circ$
2 $\times$	15.86	0.97	16.00	37.34 dB	3.95 $^\circ$
3 $\times$	15.86	0.25	64.00	30.70 dB	5.44 $^\circ$
0 $\times$	37.00	37.00	1.00	43.68 dB	1.95°
1 $\times$	37.00	9.25	4.00	39.66 dB	3.03 $^\circ$
2 $\times$	37.00	2.31	16.00	37.53 dB	3.84 $^\circ$
3 $\times$	37.00	0.58	64.00	29.78 dB	6.20 $^\circ$
0 $\times$	55.50	55.50	1.00	38.77 dB	3.09°
1 $\times$	55.50	13.88	4.00	37.71 dB	3.81 $^\circ$
2 $\times$	55.50	3.44	16.00	36.91 dB	3.87 $^\circ$
3 $\times$	55.50	0.86	64.00	31.15 dB	5.42 $^\circ$
0 $\times$	111.00	111.00	1.00	29.33 dB	8.88 $^\circ$
1 $\times$	111.00	27.75	4.00	34.98 dB	4.55 $^\circ$
2 $\times$	111.00	6.94	16.00	35.23 dB	4.31°
3 $\times$	111.00	1.73	64.00	25.74 dB	11.01 $^\circ$
1 $\times$	222.00	55.50	4.00	30.81 dB	5.82 $^\circ$
2 $\times$	222.00	13.88	16.00	31.35 dB	5.38°
3 $\times$	222.00	3.47	64.00	29.93 dB	6.16 $^\circ$
1 $\times$	444.00	111.00	4.00	26.14 dB	11.63 $^\circ$
2 $\times$	444.00	27.75	16.00	28.48 dB	6.50°
3 $\times$	444.00	6.94	64.00	25.56 dB	10.98 $^\circ$
2 $\times$	888.00	55.50	16.00	26.19 dB	8.63°
3 $\times$	888.00	13.88	64.00	26.14 dB	8.73 $^\circ$
2 $\times$	1,776.00	111.00	16.00	24.52 dB	10.07°
3 $\times$	1,776.00	27.75	64.00	24.49 dB	10.23 $^\circ$

three spatial stages to 49.78, 48.45, and 44.65 dB, respectively. This can be explained by the relatively high number of latent channels retained in this CR range by HyCASS models without spatial compression (e.g., $\Gamma = 51$ for $\text{CR} \approx 4$), which provide sufficient spectral information for accurate reconstruction by the decoder without requiring any spatial feature aggregation. For HySpecNet-11 k, spatial compression poses reconstruction challenges due to the limited spatial correlation caused by the low spatial resolution, leading to noticeably blurred reconstruction results. Interestingly, at low CRs, spatio-spectral HyCASS models tend to exhibit PSNR saturation. This behavior suggests that the high number of latent channels in such configurations contains redundancy, allowing comparable reconstruction performance at significantly higher CRs.

Second, as the CR increases beyond 32, the performance of HyCASS models that rely solely on spectral compression with zero spatial stages diminishes rapidly. In contrast, HyCASS models that incorporate spatial compression (one or more spatial stages) maintain a higher quality of reconstruction at these compression levels. This behavior indicates that with higher CRs, where spectral compression reaches saturation, the inclusion of deeper spatial hierarchies becomes increasingly important for preserving structural and spectral fidelity. In particular, this trend persists even for $\text{CRs} > 64$, where models with a higher

number of spatial stages consistently achieve better reconstruction performance. These findings highlight the increasing importance of spatial compression in highly constrained compression scenarios.

Third, reconstruction quality generally decreases with increasing CRs, dropping from 56.44 dB at $\text{CR} = 3.96$ to 41.46 dB at $\text{CR} = 808$ with the respective optimal spatial stages. This trend is consistently observed by increasing SA values, indicating that stronger compression leads to greater loss of spatial and spectral fidelity.

2) *Berlin-Urban-Gradient*: Table III shows the results of the ablation study of HyCASS on the Berlin-Urban-Gradient dataset. The results in the table indicate that for $\text{CRs} < 64$, purely spectral compression with $0 \times$ spatial stages outperforms configurations that exploit spatial redundancies. This is due to the available spectral redundancies inside the 111 spectral bands, which for low CRs yield better compression potential than spatial compression through spatial stages. Once the CR increases further, limiting the latent space to only one latent channel, the spatial stages become essential to enable effective spatio-spectral compression. Thus, for $\text{CRs} > 64$, employing two spatial stages proves advantageous compared to spectral compression. The immediate increase to two spatial stages highlights the importance of exploiting spatial redundancies in HSIs with medium spatial resolution at medium and high CRs.

TABLE IV
HYCASS RESULTS OBTAINED BY VARYING THE SPATIAL STAGES S ON THE
TEST SET OF THE MLRetSet [31] DATASET

S	CR	CR _{spec}	CR _{spat}	PSNR $\uparrow$	SA $\downarrow$
0 ×	4.01	4.01	1.00	44.86 dB	1.44°
1 ×	3.98	0.99	4.00	42.64 dB	1.85°
2 ×	3.97	0.25	16.00	42.28 dB	1.89°
3 ×	3.97	0.06	64.00	40.34 dB	2.15°
0 ×	7.85	7.85	1.00	44.89 dB	1.44°
1 ×	7.81	1.95	4.00	42.62 dB	1.85°
2 ×	7.78	0.49	16.00	42.12 dB	1.91°
3 ×	7.77	0.12	64.00	40.98 dB	2.07°
0 ×	16.04	16.04	1.00	44.86 dB	1.44°
1 ×	15.70	3.93	4.00	42.93 dB	1.79°
2 ×	15.58	0.97	16.00	42.23 dB	1.90°
3 ×	15.55	0.24	64.00	40.36 dB	2.15°
0 ×	30.75	30.75	1.00	44.79 dB	1.45°
1 ×	28.94	7.24	4.00	42.63 dB	1.85°
2 ×	28.94	1.81	16.00	42.27 dB	1.89°
3 ×	28.87	0.45	64.00	37.63 dB	2.72°
0 ×	61.50	61.50	1.00	44.23 dB	1.55°
1 ×	61.50	15.38	4.00	42.89 dB	1.78°
2 ×	61.50	3.84	16.00	42.28 dB	1.89°
3 ×	61.50	0.96	64.00	41.28 dB	2.01°
0 ×	123.00	123.00	1.00	42.93 dB	1.80°
1 ×	123.00	30.75	4.00	42.44 dB	1.86°
2 ×	123.00	7.69	16.00	42.27 dB	1.89°
3 ×	123.00	1.92	64.00	41.42 dB	1.97°
0 ×	184.50	184.50	1.00	40.98 dB	2.16°
1 ×	184.50	46.13	4.00	42.40 dB	1.84°
2 ×	184.50	11.53	16.00	42.17 dB	1.91°
3 ×	184.50	2.88	64.00	41.13 dB	2.02°
0 ×	369.00	369.00	1.00	33.55 dB	4.81°
1 ×	369.00	92.25	4.00	42.17 dB	1.90°
2 ×	369.00	23.06	16.00	42.13 dB	1.91°
3 ×	369.00	5.77	64.00	41.97 dB	1.88°
1 ×	738.00	184.50	4.00	40.43 dB	2.24°
2 ×	738.00	46.13	16.00	41.43 dB	2.02°
3 ×	738.00	11.53	64.00	41.57 dB	1.94°
1 ×	1,476.00	369.00	4.00	37.68 dB	2.83°
2 ×	1,476.00	92.25	16.00	40.80 dB	2.08°
3 ×	1,476.00	23.06	64.00	41.04 dB	2.01°

3) *MLRetSet*: Table IV shows the ablation study of HyCASS on the MLRetSet dataset. One can observe that spatio-spectral compression becomes effective primarily at CRs > 128. This behavior results from the high number of spectral bands (369) in MLRetSet, which allows spectral compression to maintain its effectiveness across a broader range of CRs before saturation. Consequently, the performance disparity between spectral and spatio-spectral compression is less pronounced at lower CRs. This suggests that spatial information plays a more significant role in MLRetSet, due to its high spatial resolution.

B. Comparison With Other Approaches

This subsection analyzes the effectiveness of HyCASS in terms of PSNR at different CRs comparing it with several traditional baselines and state-of-the-art learning-based HSI compression models on the HySpecNet-11k, the Berlin-Urban-Gradient, and the MLRetSet datasets. The comparative models include: JPEG2000; PCA; 1D-CAE [23]; SSCNet [25]; 3D-CAE [26]; HyCoT [28]; and Verdú et al.'s model [20]. Fig. 6 illustrates the corresponding rate-distortion curves, where the rate is expressed as the CR and distortion is measured as PSNR in decibel.

1) *HySpecNet-11k*: Fig. 6(a) shows that our proposed model achieves superior PSNR reconstruction quality on the HySpecNet-11k dataset (low spatial resolution) across nearly all CRs when compared to state-of-the-art learning-based models. Only the method from [20] achieves a higher PSNR for CRs between 8 and 16, despite focusing on the compression of spatial redundancies, due to its particular architectural design for this compression range. Moreover, traditional methods demonstrate superior performance at CRs < 64. JPEG2000 performs better for CRs below 16, while PCA gives higher PSNR value even for CRs up to 64. These results suggest that for CRs < 64, traditional methods remain more effective, and the increased complexity of learning-based models may offer limited benefits in this CR range. Furthermore, it shows that spectral compression is preferable to spatial compression for low CRs when the spatial resolution is low. At a CR $\approx$ 101, HyCASS with two spatial stages (spatio-spectral compression) reaches a PSNR of 46.84 dB, clearly surpassing the best-performing state-of-the-art model SSCNet (spatial compression), which achieves 43.597 dB and also the PCA baseline (spectral compression), which achieves 44.76 dB. This suggests that spatio-spectral compression is beneficial when dealing with medium CRs and low spatial resolution. Similarly, at the CR of approximately 1024, HyCASS achieves a PSNR of 41.46 dB, outperforming SSCNet and JPEG2000 that reach 40.11 dB and 35.47 dB, respectively. For CRs > 64, our results demonstrate the superior effectiveness of our proposed model over the other learning-based models and traditional approaches, due to HyCASS's adjustable spatio-spectral compression. A comparative analysis of state-of-the-art learning-based models reveals that for CR < 32, our proposed model, which exclusively performs spectral compression in this range without any spatial stages, shows limited improvement over the state of the art. This is because HyCASS is not specifically optimized for this CR range, where solely spectral compression is sufficient, resulting in performance closely similar to HyCoT, a spectral compression model. For CR > 256, HyCASS's performance converges with that of SSCNet, a spatial compression model. SSCNet naturally excels under these conditions due to its design for strong spatial compression. It is worth noting that the performance of HyCASS at CR > 256 could potentially be further enhanced by integration of additional spatial stages (e.g., beyond three), which would allow for an even more effective exploitation of spatial redundancies.

2) *Berlin-Urban-Gradient*: Fig. 6(b) presents the experimental results on the Berlin-Urban-Gradient dataset, which has a medium spatial resolution of 3.6 m. For CRs below 16, PCA demonstrates superior performance, while JPEG2000 remains competitive for CRs below 8. These observations suggest that, at low CRs, traditional methods are more effective than learning-based models, potentially due to the limited size of the dataset, which makes the training of learning-based models difficult. SSCNet performs poorly across all tested CRs, and the 3D-CAE shows limited improvement compared to that, saturating at relatively low PSNRs < 35 dB. This highlights the need for a careful design when incorporating spatial redundancies. In contrast, the 1D-CAE, which captures short-range spectral dependencies,

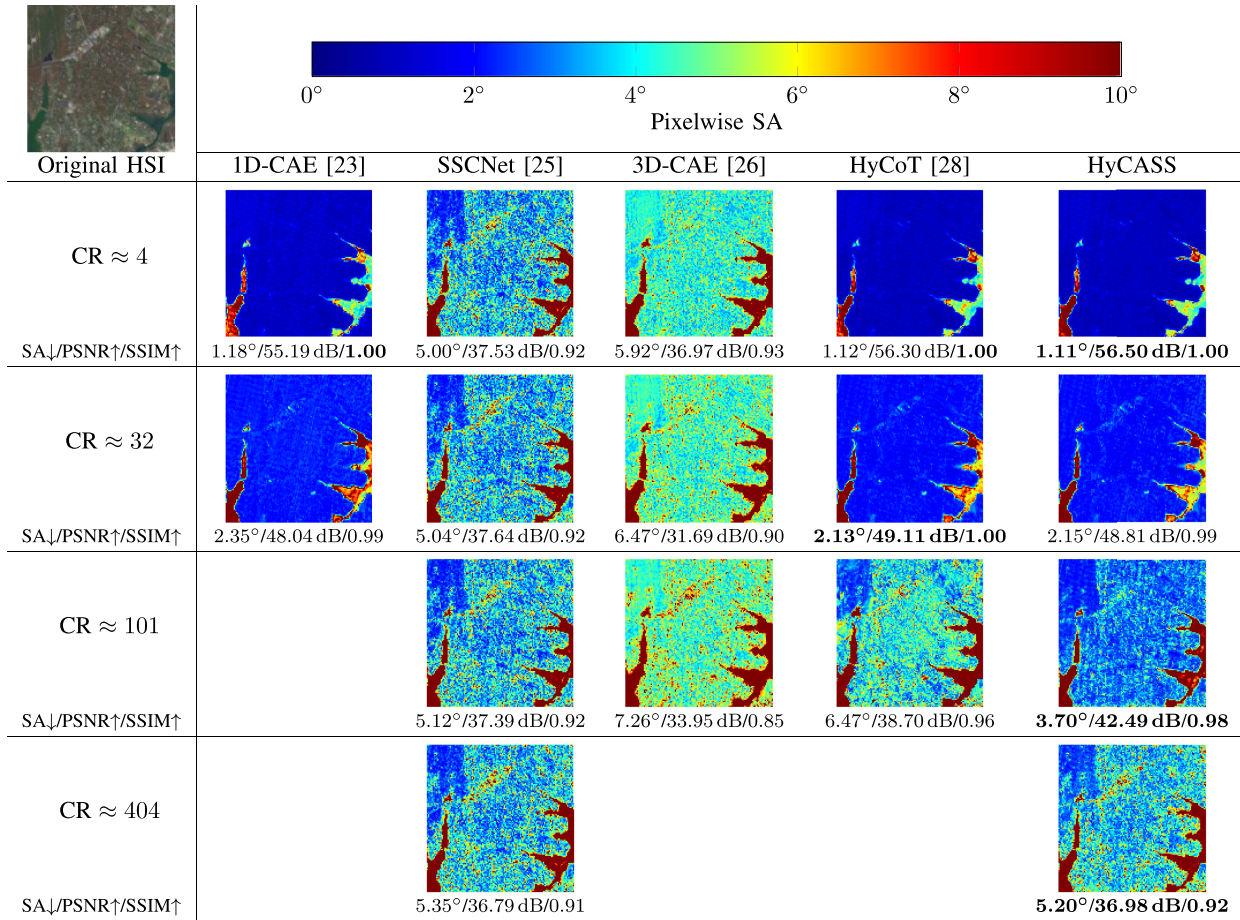


Fig. 7. SA error maps of state-of-the-art learning-based HSI compression models across three CRs, evaluated on an example image from the HySpecNet-11k [29] dataset.

outperforms HyCoT, whose reliance on long-range dependencies is less effective in this case due to the comparatively low number of only 111 spectral bands. Finally, HyCASS achieves clear benefits for CRs > 64 by leveraging two spatial stages to achieve spatio-spectral compression.

3) *MLRetSet*: Fig. 6(c) presents the rate-distortion curves on the MLRetSet dataset, from which the following observations can be made: Traditional approaches are highly effective in low-compression regimes. At CR < 64 , JPEG2000 performs particularly well, surpassing all learning-based models and the PCA baseline. The reconstruction quality achieved by the spectral learning-based models 1D-CAE, HyCoT, and HyCASS with zero spatial stages shows only minor variation, indicating comparable performance for long-range and short-range spectral redundancy compression. SSCNet and 3D-CAE, which incorporate spatial compression, perform worse than the spectral compression models, especially for CR < 64 . This suggests that, when sufficient bitrate is available, exploiting spectral redundancies is more straightforward and yields better compression performance than incorporating spatial information. HyCASS achieves comparable performance to learning-based spectral compression models at CR < 128 , while at CR > 128 , it demonstrates advantages thanks to its adjustable design by integrating spatial compression.

C. Visual Analysis

For a qualitative evaluation, the reconstruction outputs of the considered learning-based compression models are visually compared under varying CRs on two exemplary HSIs, which include different land cover types, in Figs. 7 and 8. The error maps show the reconstruction error (derived from the SA for each pixel) across several representative CRs. This provides a detailed visual assessment of the spatial distribution of reconstruction errors. Each case also reports the corresponding CR, the overall SA, PSNR, and SSIM for comprehensive comparison.

Fig. 7 presents the error maps for a reconstructed HySpecNet-11k image, covering mostly urban, water, and forest areas. At CR ≈ 4 , the spectral compression models 1D-CAE, HyCoT, and HyCASS with zero spatial stages achieve relatively low SA errors of slightly above 1° across most of the scene, as also indicated by their high overall PSNR values exceeding 55 dB and an SSIM of 1.00. Urban areas are reconstructed with high precision, showing SA $< 2^\circ$, while water regions pose more difficulty, with errors exceeding SA $> 5^\circ$. In contrast, SSCNet and 3D-CAE achieve lower PSNR values of approximately 37 dB and also the SSIM is decreasing to 0.92 and 0.93, respectively. Consequently, their error maps contain considerable noise in the urban regions. Similar to spectral compression models, these models also face difficulties in reconstructing water areas, whereas the

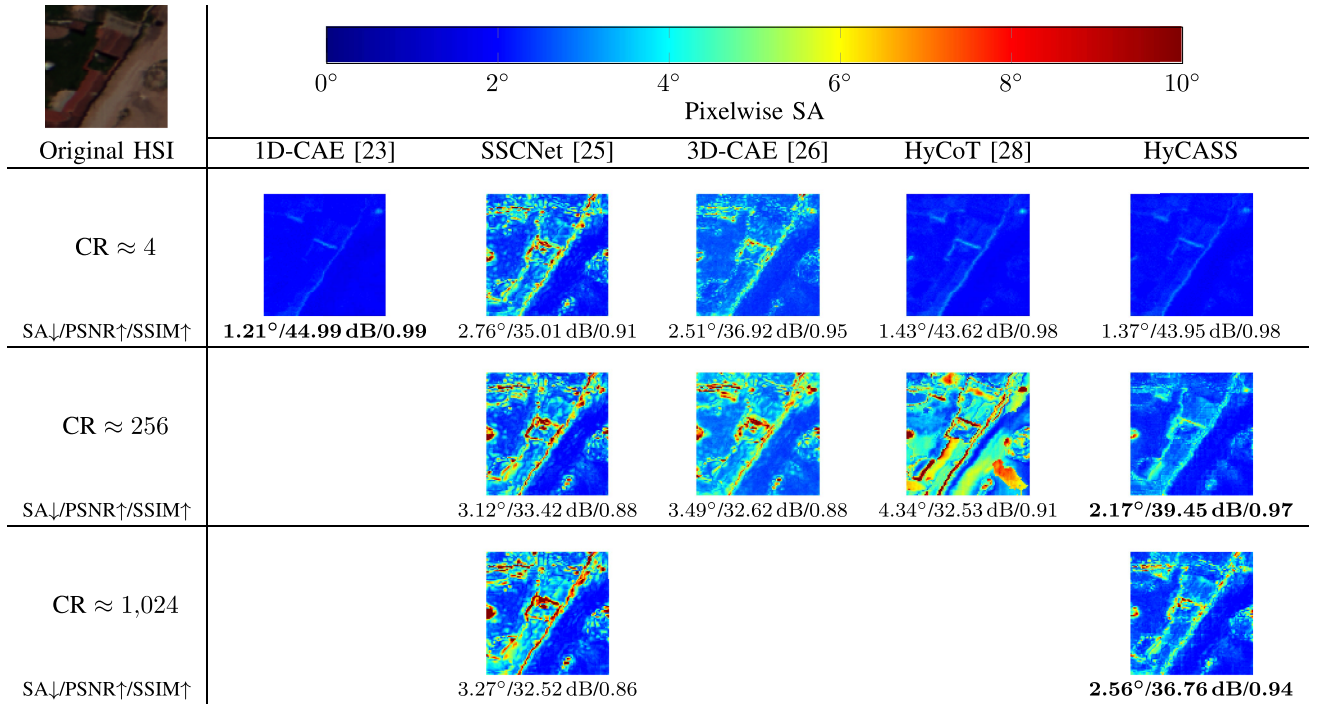


Fig. 8. SA error maps of state-of-the-art learning-based HSI compression models across three CRs, evaluated on an example image from the MLRetSet [31] dataset.

forest region in the top-left shows the highest reconstruction fidelity. This suggests that at such low CR, spatial compression adversely affects reconstruction quality, limiting the model's ability to preserve fine spectral details of each pixel due to the spatial downsampling. At $CR \approx 32$, distortion increases for most methods. SSCNet shows a slight improvement in reconstruction quality compared to its performance at $CR \approx 4$, although overall quality remains low. In addition, the models 1D-CAE, HyCoT, and HyCASS exhibit increased distortion in urban regions, with notably elevated SA values observed in the industrial area situated at the top center of the HSI. At $CR \approx 101$, reconstruction degradation becomes more visible. While SSCNet and 3D-CAE exhibit minimal differences to lower CRs, the spectral compression model HyCoT exhibits substantial errors in urban and water regions, however it maintains relatively strong performance in forested areas. In contrast, HyCASS, employing two spatial stages at this CR, exhibits increased error in urban regions but it achieves the highest overall SA, PSNR, and SSIM. Further degradation can be observed at $CR \approx 404$.

Fig. 8 shows the error maps of a reconstructed MLRetSet image, which includes mainly buildings, sand roads, and grassland areas, for three different CRs. At $CR \approx 4$, the spectral compression models 1D-CAE, HyCoT, and HyCASS with zero spatial stages exhibit reconstruction errors primarily in the shadow regions of buildings while urban areas and grassland are reconstructed with high accuracy ($SA < 2^\circ$). SSCNet and 3D-CAE exhibit their major reconstruction errors along the object boundaries of buildings due to their spatial compression. At $CR \approx 256$, HyCASS with a single spatial stage demonstrates

an increase in SA for urban regions, while sand roads and grassland areas remain comparatively stable. In contrast, SSCNet and 3D-CAE display localized degradation that extends further into the buildings, while HyCoT yields a more homogeneous error distribution across the entire HSI, which shows the limit of spectral compression models when reaching higher CRs. At $CR \approx 1024$, reconstruction quality continues to decrease in the building areas, while grassland and the sand road areas retain a low reconstruction error.

The qualitative results indicate that learning-based models struggle to reconstruct water, shadow, and urban regions. Urban areas become more challenging to reconstruct at higher CRs and when spatial compression is applied. In contrast, forestry regions and vegetated areas are consistently reconstructed with the highest quality, due to their frequent representation and homogeneous spatial and spectral characteristics in the training data, which facilitate accurate learning and reconstruction. Overall, HyCASS consistently achieves an effective tradeoff between CR and reconstruction fidelity across a wide range of CRs and land cover types.

V. CONCLUSION

In this article, we have introduced HyCASS, a novel learning-based HSI compression model designed for adjustable spatio-spectral HSI compression. To this end, the proposed model employs six modules: a spectral encoder module; a spatial encoder module; a CR adapter encoder module; a CR adapter decoder module; a spatial decoder module; and a spectral decoder module.

Our model accomplishes the following:

- 1) spectral feature extraction (realized within the spectral encoder and decoder modules);
- 2) spatial compression with variable stages (realized within the spatial encoder and decoder modules); and
- 3) spectral compression with variable output channels (realized within the CR adapter encoder and CR decoder modules).

Unlike existing learning-based HSI compression models, HyCASS provides flexible control over the tradeoff between spectral and spatial compression through its modular design. We have conducted extensive experiments on three benchmark datasets, including ablation studies, comparisons with state-of-the-art methods, and visual analyses of reconstruction errors. Our results demonstrate the effectiveness of HyCASS, and reveal how the balance between spectral and spatial compression affects reconstruction fidelity across different CRs and spatial resolutions. Our findings confirm the importance of adjustable spatio-spectral compression in addressing the diverse characteristics of hyperspectral data. It is worth emphasizing that with the continuous growth of hyperspectral data archives, spatio-spectral learning-based HSIs compression is becoming increasingly important, as it enables significantly higher CRs. In this context, the proposed model offers a promising solution for efficient and flexible HSI compression. Based on our analyses, we have also derived a guideline to select the tradeoff between spectral or spatial compression depending on the spatial resolution and overall CR as follows.

- 1) *For HSI data with low spatial resolution:* Use spectral compression at low CRs; spatio-spectral compression with greater spectral emphasis at medium CRs; and spatio-spectral compression with greater spatial emphasis at high CRs.
- 2) *For HSI data with medium spatial resolution:* Use spectral compression at low CRs; and balanced spatio-spectral compression at medium and high CRs.
- 3) *For HSI data with high spatial resolution:* Use spectral compression at low CRs; and spatio-spectral compression with greater spatial emphasis at medium and high CRs.

As a final remark, we would like to note that the development of foundation models (FMs) has attracted great attention in RS. FMs are usually pretrained on large-scale datasets, and then, fine-tuned for specific downstream tasks such as image classification, segmentation, or change detection. We believe that leveraging the generalization capabilities of FMs for HSI compression can lead to a more robust and effective compression performance. As a future work, we plan to investigate the use of FMs as a backbone for HSI compression to improve reconstruction fidelity and enable adaptation across diverse sensor and acquisition conditions.

REFERENCES

- [1] M.-F. Cheng, A. Mukundan, R. Karmakar, M. A. E. Valappil, J. Jouhar, and H.-C. Wang, "Modern trends and recent applications of hyperspectral imaging: A review," *Technologies*, vol. 13, no. 5, 2025, Art. no. 170.
- [2] F.-C. Lin, Y.-S. Shiu, P.-J. Wang, U.-H. Wang, J.-S. Lai, and Y.-C. Chuang, "A model for forest type identification and forest regeneration monitoring based on deep learning and hyperspectral imagery," *Ecological Inform.*, vol. 80, 2024, Art. no. 102507.
- [3] A. Fabbretto et al., "Tracking water quality and macrophyte changes in Lake Trasimeno (Italy) from spaceborne hyperspectral imagery," *Remote Sens.*, vol. 16, no. 10, 2024, Art. no. 1704.
- [4] D. Spiller, A. Carbone, S. Amici, K. Thangavel, R. Sabatini, and G. Laneve, "Wildfire detection using convolutional neural networks and PRISMA hyperspectral imagery: A spatial-spectral analysis," *Remote Sens.*, vol. 15, no. 19, 2023, Art. no. 4855.
- [5] I. Masari, G. Moser, and S. B. Serpico, "Manifold learning and deep generative networks for heterogeneous change detection from hyperspectral and synthetic aperture radar images," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, 2024, Art. no. 5500105.
- [6] Y. Liu et al., "Utilizing UAV-based hyperspectral remote sensing combined with various agronomic traits to monitor potato growth and estimate yield," *Comput. Electron. Agriculture*, vol. 231, 2025, Art. no. 109984.
- [7] M. Balsi, M. Moroni, and S. Bouchelaghem, "Plastic litter detection in the environment using hyperspectral aerial remote sensing and machine learning," *Remote Sens.*, vol. 17, no. 5, 2025, Art. no. 938.
- [8] C. Gomes et al., "Lossy neural compression for geospatial analytics: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 13, no. 3, pp. 97–135, Sep. 2025.
- [9] A. Altamimi and B. Ben Youssef, "Lossless and near-lossless compression algorithms for remotely sensed hyperspectral images," *Entropy*, vol. 26, no. 4, 2024, Art. no. 316.
- [10] Y. Dua, V. Kumar, and R. S. Singh, "Comprehensive review of hyperspectral image compression algorithms," *Opt. Eng.*, vol. 59, no. 9, 2020, Art. no. 090902.
- [11] J. Mielikainen and P. Toivanen, "Clustered DPCM for the lossless compression of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 12, pp. 2943–2946, Dec. 2003.
- [12] M. Hernández-Cabronero et al., "The CCSDS 123.0-B-2 "low-complexity lossless and near-lossless multispectral and hyperspectral image compression" standard: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 102–119, Dec. 2021.
- [13] Q. Du and J. E. Fowler, "Hyperspectral image compression using JPEG2000 and principal component analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 2, pp. 201–205, Apr. 2007.
- [14] R. J. Yadav and M. Nagmode, "Compression of hyperspectral image using PCA-DCT technology," *Innovations Electron. Commun. Eng.*, vol. 7, pp. 269–277, 2018.
- [15] P. L. Dragotti, G. Poggi, and A. R. Ragozini, "Compression of multispectral images by three-dimensional SPIHT algorithm," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 1, pp. 416–428, Jan. 2000.
- [16] X. Tang and W. A. Pearlman, "Three-dimensional wavelet-based compression of hyperspectral images," *Hyperspectral Data Compression*, Boston, MA, USA: Springer US, 2006, pp. 273–308.
- [17] D. Valsesia, T. Bianchi, and E. Magli, "Onboard deep lossless and near-lossless predictive coding of hyperspectral images with line-based attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5532714.
- [18] M. H. P. Fuchs, A. P. Byju, A. Walda, B. Rasti, and B. Demir, "Generative adversarial networks for spatio-spectral compression of hyperspectral images," in *Proc. IEEE Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens.*, 2024, pp. 1–5.
- [19] F. Mentzer, G. D. Toderici, M. Tschannen, and E. Agustsson, "High-fidelity generative image compression," *Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 11913–11924, 2020.
- [20] S. Mijares i Verdú, J. Ballé, V. Laparra, J. Bartrina-Rapesta, M. Hernández-Cabronero, and J. Serra-Sagrístà, "A scalable reduced-complexity compression of hyperspectral remote sensing images using deep learning," *Remote Sens.*, vol. 15, no. 18, 2023, Art. no. 4422.
- [21] S. Rezasoltani and F. Z. Qureshi, "Hyperspectral image compression using sampling and implicit neural representations," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2024, Art. no. 5500712.
- [22] N. Zhao, T. Pan, Z. Li, E. Chen, and L. Zhang, "The paradigm shift in hyperspectral image compression: A neural video representation methodology," *Remote Sens.*, vol. 17, no. 4, 2025, Art. no. 679.
- [23] J. Kuester, W. Gross, and W. Middelmann, "1D-convolutional autoencoder based hyperspectral data compression," *Int. Arch. Photogrammetry, Remote Sens. Spatial Inf. Sci.*, vol. 43, pp. 15–21, 2021.
- [24] J. Kuester, W. Gross, S. Schreiner, W. Middelmann, and M. Heizmann, "Adaptive two-stage multisensor convolutional autoencoder model for lossy compression of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5530022.

- [25] R. La Grassa, C. Re, G. Cremonese, and I. Gallo, "Hyperspectral data compression using fully convolutional autoencoder," *Remote Sens.*, vol. 14, no. 10, 2022, Art. no. 2472.
- [26] Y. Chong, L. Chen, and S. Pan, "End-to-end joint spectral-spatial compression and reconstruction of hyperspectral images using a 3D convolutional autoencoder," *J. Electron. Imag.*, vol. 30, no. 4, 2021, Art. no. 041403.
- [27] N. Sprengel, M. H. P. Fuchs, and B. Demir, "Learning-based hyperspectral image compression using a spatio-spectral approach," EGU General Assembly, 2024.
- [28] M. H. P. Fuchs, B. Rasti, and B. Demir, "HyCoT: A transformer-based autoencoder for hyperspectral image compression," in *Proc. IEEE Workshop Hyperspectral Imag. Signal Process., Evol. Remote Sens.*, 2024, pp. 1–5.
- [29] M. H. P. Fuchs and B. Demir, "HySpecNet-11k: A large-scale hyperspectral dataset for benchmarking learning-based hyperspectral image compression methods," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, 2023, pp. 1779–1782.
- [30] A. Okujeni, S. van der Linden, and P. Hostert, "Berlin-urban-gradient dataset 2009—An enmap preparatory flight campaign," *GFZ Data Serv.*, 2016.
- [31] F. Ömrüüzun, Y. Yardımcı Çetin, U. M. Leloğlu, and B. Demir, "A novel semantic content-based retrieval system for hyperspectral remote sensing imagery," *Remote Sens.*, vol. 16, no. 8, 2024, Art. no. 1462.
- [32] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 10012–10022.
- [33] M. Lu, P. Guo, H. Shi, C. Cao, and Z. Ma, "Transformer-based image compression," in *Proc. Data Compression Conf.*, 2022, Art. no. 469.
- [34] L. Guanter et al., "The EnMAP spaceborne imaging spectroscopy mission for earth observation," *Remote Sens.*, vol. 7, no. 7, pp. 8830–8857, 2015.
- [35] T. Cocks, R. Jenssen, A. Stewart, I. Wilson, and T. Shields, "The HyMap airborne hyperspectral sensor: The system, calibration and performance," in *Proc. EARSeL Workshop Imag. Spectrosc.*, 1998, pp. 37–42.
- [36] J. Bégaïnt, F. Racapé, S. Feltman, and A. Pushparaja, "CompressAI: A PyTorch library and evaluation platform for end-to-end compression research," 2020, *arXiv:2011.03029*.
- [37] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.



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